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Education

Ph.D. in Computer Science , University of Maryland – College Park, US	Jan 2022 – present
• Field: Theoretical Computer Science • Designing algorithms for fundamental graph problems and optimizing submodular functions in dynamic settings.	
M.Sc. in Computer Engineering , Sharif University of Technology – Tehran, Iran	Sept 2019 – Dec 2021
• Field: Algorithms and Computation • Studied geometric graph problems in the massively parallel computation (MPC) model.	
B.Sc. in Computer Engineering , Sharif University of Technology – Tehran, Iran	Sept 2015 – Aug 2019

Awards

Best Paper Award 🔗, Symposium on Foundations of Computer Science (FOCS)	Dec 2025
Asia West Champion , International Collegiate Programming Contest (ICPC) World Finals	Apr 2018
14th Place , ACM International Collegiate Programming Contest (ACM-ICPC) World Finals	May 2016
Bronze Medal 🔗, International Olympiad in Informatics (IOI)	July 2015
Gold Medal , Asia-Pacific Informatics Olympiad (APIO)	May 2015
Gold Medal , Iranian National Olympiad in Informatics	Sept 2014

Experience

Software Engineer Intern , Axiom 🔗 – New York, US	June 2025 – Aug 2025
• Designed and implemented novel ZK circuits, and optimized existing ones to efficiently support program instructions in zkVM.	
Software Engineer , Balad (Map and Navigation App) – Tehran, Iran	Sept 2018 – Nov 2021
• Enhanced service stability and scalability and optimized inter-service communication as part of the infrastructure team. • Developed and took ownership of Balad's core service for managing and providing location-specific data. • Led a diverse team of 10 engineers to collect and clean location-specific data, and to refine the service design using microservices. • Primarily coded in Python, with additional experience in C++, Java, Go, and various other tools.	
Research Intern , Max Planck Institute for Informatics – Saarbrücken, Germany	July 2019 – Aug 2019
• Designed and benchmarked an energy-preserving online scheduling algorithm.	
Software Engineer Intern , Balad (Map and Navigation App) – Tehran, Iran	June 2018 – Aug 2018
• Designed and implemented a novel algorithm for optimal routing of camera-equipped cars, ensuring comprehensive street coverage in Tehran within the minimum time.	

Publications

(All papers use **alphabetical** author ordering.)

Breaking a Long-Standing Barrier: $2-\epsilon$ Approximation for Steiner Forest 🔗	Dec 2025
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi	
66th Annual Symposium on Foundations of Computer Science (FOCS 2025, best paper award).	
Dynamic Diameter in High-Dimensions against Adaptive Adversary and Beyond 🔗	Dec 2025
Kiarash Banihashem, Jeff Giliberti, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh	
39th Conference on Neural Information Processing Systems (NeurIPS 2025).	
Non-monotone Submodular Optimization: p-Matchoid Constraints and Fully Dynamic Setting 🔗	Dec 2025
Kiarash Banihashem, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh	
39th Conference on Neural Information Processing Systems (NeurIPS 2025).	
Dynamic Algorithms for Submodular Matching 🔗	July 2025
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh	
52nd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2025).	
Prize-Collecting Forest with Submodular Penalties: Improved Approximation 🔗	June 2025
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi	
26th Conference on Integer Programming and Combinatorial Optimization (IPCO 2025).	

A Dynamic Algorithm for Weighted Submodular Cover Problem	July 2024
Kiarash Banihashem, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh 41st International Conference on Machine Learning (ICML 2024, accepted for oral presentation)	
Prize-Collecting Steiner Tree: A 1.79 Approximation	June 2024
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi 56th ACM Symposium on Theory of Computing (STOC 2024).	
2-Approximation for Prize-Collecting Steiner Forest	Jan 2024
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi Journal of the ACM (JACM 2025). ACM-SIAM Symposium on Discrete Algorithms (SODA 2024, selected for special issue).	
Dynamic Algorithms for Matroid Submodular Maximization	Jan 2024
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh ACM-SIAM Symposium on Discrete Algorithms (SODA 2024).	
Dynamic Non-monotone Submodular Maximization	Dec 2023
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh 37th Conference on Neural Information Processing Systems (NeurIPS 2023).	
Dynamic Constrained Submodular Optimization with Polylogarithmic Update Time	July 2023
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh 40th International Conference on Machine Learning (ICML 2023)	
A Novel Prediction Setup for Online Speed-Scaling	June 2022
Antonios Antoniadis, Peyman Jabbarzade, Golnoosh Shahkarami 18th Scandinavian Symposium and Workshops on Algorithm Theory (SWAT 2022)	

Academic Talks

The University of Michigan , Breaking a Long-Standing Barrier: $2-\epsilon$ Approximation for Steiner Forest	Jan 2026
Johns Hopkins University , Breaking a Long-Standing Barrier: $2-\epsilon$ Approximation for Steiner Forest	Oct 2025
The University of Washington , Breaking a Long-Standing Barrier: $2-\epsilon$ Approximation for Steiner Forest	June 2025
The University of Maryland , Breaking a Long-Standing Barrier: $2-\epsilon$ Approximation for Steiner Forest	May 2025
ICERM: Fusing Theory and Practice of Graph Algorithms , Prize-Collecting Framework	Feb 2025
SODA 2024 , 2-Approximation for Prize-Collecting Steiner Forest	Jan 2024
The University of Maryland , 2-Approximation for Prize-Collecting Steiner Forest	Sept 2023

Volunteer Experience

Paper Reviewing	2020 – present
- Served as a Program Committee member for AAAI 2026 and as a Junior Program Committee member for SPAA 2025. - Served as a reviewer for Algorithmica journal and for conferences including SODA, NeurIPS, ICML, ICLR, IPCO, ICALP, SPAA, SOSA, AAAI, AISTATS, APPROX, ESA, DISC, and ALGOSENSORS.	
Coaching	2020 – 2023
- Coached the University of Maryland's team for ICPC 2023, leading them to a silver medal at the North America Championship and advancing to the world final. [1, 2] - Led Iran's national team for IOI 2020, achieving 3 gold and 1 silver medals, the best result for Iran in IOI. - Coached Sharif University of Technology's team in the ICPC 2020 World Finals.	
Scientific Committee	2017 – 2024
- Served as a member of the Host Scientific Committee (HSC) for IOI 2017 and IOI 2019. - Authored problems for several programming competitions, including "Rectangles" for IOI 2019. - Judge for the ICPC North America Championship 2024. - Member of the Iranian National Olympiad in Informatics Scientific Committee over six years. - Contributed to the preparation of the ICPC Tehran Regional Contest over six years.	
Teaching	2015 – 2024
- Teaching Assistant at the University of Maryland for courses in Algorithms, Advanced Data Structures, and Computational Genomics. - Teaching Assistant at Sharif University of Technology for courses in Design Algorithms and Discrete Structures. - Prepared high school students for IOI.	